Section A: Pure Mathematics [35 marks]

1 Solve the simultaneous equations

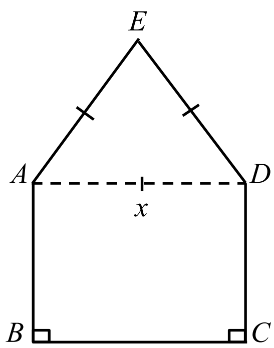
$$2^x = 4(8^y),$$

$$\log_5 x - \log_5(x - 4y) = 1. \qquad [4]$$

2 **(i)** Sketch the graphs of $y = e^x + 1$ and $y = ex + 1$ on the same axes, giving the equations of any asymptotes and the exact coordinates of the point of intersection between the two graphs. [3]

(ii) Find the exact area of the finite region bounded by the graphs in part **(i)** and the y-axis. [3]

3 A window has the shape of a rectangle *ABCD* joined to an equilateral triangle *ADE*. It is given that $AD = x$ m, the total perimeter *ABCDEA* of the window is 24 m and the enclosed area is S m^2.



(i) Show that $S = 12x + \left(\frac{\sqrt{3}}{4} - \frac{3}{2}\right)x^2$. [3]

(ii) Given that x can vary, use differentiation to find the stationary value of S and determine whether it is a maximum or minimum. Give your answer correct to 2 decimal places. [4]

4 A curve C has equation $y = \dfrac{1}{x} - \dfrac{5}{x^2}$.

(i) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ for exactly one value of x and find the corresponding value of y. [3]

(ii) Sketch C, indicating clearly the equations of any asymptotes, and the coordinates of any intersections with the axes. [3]

Deduce the range of values of k such that the equation

$$\frac{x-5}{x^2} - k = 0$$

has no real roots. [1]

(iii) Find the equation of the normal to C at the point where $x = 5$. [2]

5 **(i)** Differentiate

(a) $\dfrac{3}{5-x}$, [2]

(b) $\ln(2x^2+4)^3$. [2]

(ii) Without using a calculator, find the exact value of $\displaystyle\int_1^8 \left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}\right)^2 \mathrm{d}x$, simplifying your answer. [5]

Section B: Statistics [60 marks]

6 A sports club has 400 members and 3 sections – Badminton, Swimming and Tennis. The table below shows the number of males and females in each section.

	Male	Female
Badminton	105	60
Swimming	90	45
Tennis	45	55
	240	160

The names of all the members are recorded in alphabetical order. A sample of 80 members is to be chosen to take part in a survey to find out their opinions of the sports facilities offered by the club.

(i) Describe briefly how the sample could be chosen using systematic sampling. [2]

(ii) Give one disadvantage of using systematic sampling in this case. [1]

(iii) Describe an alternative sampling method which would be better in this case. [2]

7 **(a)** Two events X and Y are such that $\mathrm{P}(X)=\frac{1}{3}$, $\mathrm{P}(Y)=\frac{1}{2}$, $\mathrm{P}(X' \text{«} Y)=\frac{1}{2}$. Find

(i) $\mathrm{P}(X \text{«} Y)$, and hence state the relationship between X and Y, [2]

(ii) the value of $\mathrm{P}(X' \text{«} Y')$. [2]

(b) A boy goes to school either by bus or on foot. On a day he goes to school by bus, then the probability that he goes by bus the next day is $\frac{7}{10}$; if he walks to school, then the probability that he goes by bus the next day is $\frac{2}{5}$.

On a particular Tuesday, the boy is equally likely to walk or take a bus to school. With the help of a tree diagram, or otherwise, find the probability that he will go to school by bus on Thursday of that week, given that he walks to school that Tuesday. [4]

Suppose he walks to school on both Tuesday and Thursday of that week. Find the probability that he also walks to school on Wednesday. [2]

8 Two machines A and B produce plastic circular discs. The diameters of the discs produced by both machines are independent of each other.

Discs made by machine A have diameters following a normal distribution with mean 8.95 cm and standard deviation σ cm. It is known that 1.6% of the discs made by machine A have diameters more than 9.02 cm,

(i) find the value of σ, leaving your answer correct to 3 decimal places. [4]

Discs made by machine B have diameters with mean 9 cm and standard deviation 0.15 cm .

(ii) Calculate the probability that the average diameter of a random sample of 50 discs made by machine B is more than 9.05 cm. [3]

(iii) Assuming the diameters of the discs made by machine B follow a normal distribution, calculate the probability that the total diameter of 3 randomly chosen discs made by machine A is less than three times the diameter of a randomly chosen disc made by machine B. [3]

9 A supermarket manager is investigating the amount of time that customers spend shopping in the store per week. The times, x hours, of a random sample of 100 customers are summarised by $\sum x = 558.2$, $\sum x^2 = 4027.89$. Calculate the unbiased estimates of the population mean and variance. [2]

(i) Assuming a normal distribution, test at the 5% significance level whether the mean time spent shopping by customers, μ hours, is more than 5 hours. [4]

(ii) State, giving reason, whether the conclusion of the test in part **(i)** would be valid in each of the following situation:

(a) if it could no longer be assumed that times spent shopping has a normal distribution. [1]

(b) if the random sample of 100 customers is taken at weekends. [1]

(iii) Find, to 1 decimal place, the least value of μ_0, for which the alternative hypothesis $\mu > \mu_0$ would not be accepted at the 10% significance level. [3]

10 A small airline company has an aircraft with 45 passenger seats and 50 tickets are to be sold for each flight. On average, for every 100 tickets sold, 17 passengers will fail to appear. Let X be the number of passengers who turn up for a randomly chosen flight.

(i) State the distribution of X, giving one assumption, in the context of this question, you must make for it to be appropriate. [2]

(ii) Calculate the probability that in a randomly chosen flight there will be passengers without seats. [2]

(iii) Find the probability that out of 15 randomly chosen flights, at least 1 but at most 4 flights will have passengers without seats. [3]

(iv) There are 100 flights in a month. Using a suitable approximation, find the probability that in a month more than 5 but not exceeding 12 flights will have passengers without seats. [5]

11 An experiment to determine the effect of an organic liquid fertilizer on crop yield was carried out. A field was divided into 10 plots of equal area. Five different levels of concentration of fertiliser were used, each level being applied to two of the plots using the same quantity of liquid for each plot. The table below shows the concentration of fertilisers, x kg per litre, and the crop yield, y kg, for each plot.

x	0.0	0.0	0.5	0.5	1.0	1.0	1.5	1.5	2.0	2.0
y	6.8	7.6	7.6	8.4	8.6	9.4	9.5	10.4	10.0	11.4

(i) Draw a scatter diagram for these data as shown in your calculator. [1]

(ii) Calculate the product moment correlation coefficient between x and y and comment on its value in the context of this question. [3]

(iii) Find the equation of the regression line of y on x in the form $y = mx + c$. Give interpretations of the meaning of the two values m and c, in terms of crop yield and concentration of fertiliser. [4]

(iv) Comment, without further calculation, on the suitability of using the line found in part **(ii)** to estimate the crop yield from one of the plots when a concentration of 2.5 kg per litre is used. [1]

Use the appropriate regression line to estimate the concentration of fertiliser which would have produced a crop yield of 10.0 kg from one of the plots. Explain your choice of line and comment on the reliability of the estimate. [3]